House Price Prediction – Project Summary

Project Overview

The objective of this project was to build a machine learning model that predicts house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking availability, air conditioning, and location. The project involved data preprocessing, exploratory data analysis, model training, evaluation, and visualization to identify the factors that most influence house prices.

Data Preparation

The Housing dataset was loaded using Pandas and examined for missing values and duplicate records. Categorical features such as main road access, guest room availability, basement, air conditioning, preferred area, and furnishing status were converted into numerical form using one-hot encoding. The dataset was then prepared for machine learning model training.

Model Development

Two regression models were developed and compared:

- Linear Regression
- Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio. Both models were trained on the training data and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Key Findings

The analysis revealed that the most influential factors affecting house prices are:

- Area of the house
- Number of bathrooms
- Parking availability
- Number of stories
- Air conditioning
- Preferred location (prefarea)

Among these factors, house area showed the strongest relationship with price.

Model Performance

The Random Forest Regressor outperformed the Linear Regression model and produced more accurate predictions. The model was able to estimate house prices effectively, with predicted values closely matching actual prices for most properties.

Visualizations Created

The project included the following visualizations:

1. House Price Distribution Histogram
2. Correlation Heatmap
3. Actual vs Predicted Price Scatter Plot
4. Area vs Price Scatter Plot

These charts helped identify relationships between property features and house prices.

Conclusion and Recommendation

The project demonstrates that both property size and location significantly impact house prices. Amenities such as air conditioning and parking also contribute to higher property values. Based on these findings, real estate businesses should focus on properties located in preferred areas and highlight key amenities when marketing homes. Using machine learning-based price prediction can help improve pricing accuracy and support better decision-making in the real estate industry.